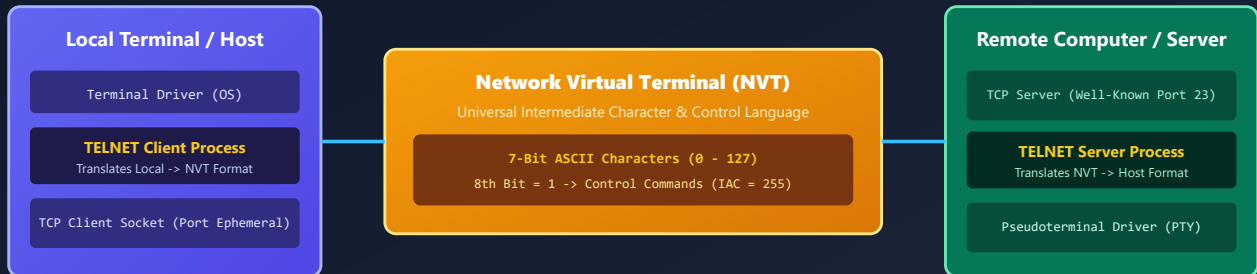


Module 03: Remote Login & TELNET Architecture

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TELNET PROTOCOL & REMOTE LOGIN ARCHITECTURE

Local vs Remote Login, Network Virtual Terminal (NVT), Option Negotiation & SSH Comparison



1. Local vs Remote Login

Local Login: Keystrokes handled by terminal driver directly to local OS shell. • **Remote Login:** Local terminal behaves as terminal for remote time-sharing host. • Transparent keystroke transmission over TCP. 2. NVT & Operating Modes • **Default Mode:** Half-duplex character-at-a-time transmission with remote echoing. • **Character Mode:** 1 char sent per packet. • **Line Mode:** Lines edited locally and transmitted in bulk on pressing ENTER. 3. TELNET vs SSH Security • **TELNET (Port 23):** Cleartext transmission! Passwords & root commands visible to packet sniffers. • **SSH - Secure Shell (Port 22):** Encrypted session tunnel using public keys and AES. Replaced TELNET globally.

1. Remote Login Paradigm

Remote login allows a terminal user on a local system to log in to and execute commands on a remote timesharing system across an IP internetwork.

The Role of Network Virtual Terminal (NVT)

To overcome terminal and keyboard heterogeneity across operating systems, TELNET maps local terminal characteristics to a universal intermediate representation called **NVT**.

2. TELNET Control Commands & Option Negotiation

Command	Byte Code	Semantic Action
IAC	255 (0xFF)	Interpret As Command (signals that subsequent byte is a TELNET control token)
WILL	251	Sender offers to initiate or enable a specific terminal capability
DO	253	Receiver requests sender to activate a specific terminal feature
WONT / DONT	252 / 254	Negative negotiation (refusal to enable requested feature)

3. TELNET vs SSH Security Analysis

Property	TELNET (Port 23)	SSH (Port 22)
Transport Security	Plaintext stream without authentication integrity	Encrypted over cryptographic tunnel with host verification
Threat Exposure	Vulnerable to sniffing and credentials interception	Immune to passive wiretapping and active session hijacking